Alexander Hall

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Purdue University, Pursuing B.S. Mechatronics — Expected Graduation: 2030
Carlmont High School (GPA 3.87/4.08) — Class of 2026

Skills

Programming: C/C++(wip), Python, Java, C#, Bash, Gamedev, Networking Basics, File Operations
Design/CAD: OnShape, Solidworks, NX(wip), Arduinos, CAN Bus wiring, Motor Controllers
Machine Tools: Band/Chop/Miter/Jig Saw, Drill Press, 3d Printing, Crimping, Soldering, Jointer/Planer

Coursework

College of San Mateo + De Anza: Comp. Networking, Solidworks I CAD, x86 Assembly and Architecture
Purdue: Materials and Processes II (Polymers)

Work Experience

TheCoderSchool San Mateo (Part-time) — August 2024 - July 2026

- Introduced and retained several students into learning programming and self-confidence.
- Improved morale by working to introduce weekly employee-selected food items.

Belmont IT Department (Part-time) — June 2023

- Set up a fleet of 70 desktop and laptop windows computers.
- Troubleshooted ID card records, account permissions, and Patrol car software.

Program/Project Experience

FRC Robotics (Programming Lead, Match Technician) — August 2022 - June 2026

- Led programming team, expanded consistent members from 4 to 12.
- Cleared a multi-year website update backlog, established next-of-kin supervision.
- Designed mechanical components (climbing mechanism, camera mounts) for team evaluation.
- Repaired and re-programmed robots during inter-match transport windows.

VEX IQ/V5 Robotics (Driver and Main Robot Designer) — August 2016 - June 2021

- Attended multiple state competitions and scored 50th place in world championships.

American Rocketry Challenge (Team Member of 3) — 2026

- Designed, a three-foot solid-fuel rocket aiming for 750 feet. Wired altimeters and gunpowder separation charges into the rocket.

Personal Projects

Utmp	Airsoft Replacement Internals	NHRL Battlebot (in progress)
Command-line youtube music and soundcloud app Bash, Multi-OS dependency downloader	3d-printed airsoft pistol parts Rapid iterations over several weeks Strength-focused parts constrained by size	PLA-only RC circuit robots Teaching teammates CAD software

Hobbies: HEMA/Swordfighting, MMA, Sewing

Github: FriedLongJohns